Geostatistical Prediction/Simulation of Point Values From Areal Data

P. C. Kyriakidis and E.-H. Yoo

Department of Geography, University of California Santa Barbara,
Ellison Hall 5710, Santa Barbara, CA 93106-4060, U.S.A.
Telephone: +1 (805) 893-2266
FAX: +1 (805) 893-3146
Email: phaedon@geog.ucsb.edu

Abstract

The spatial prediction/simulation of point values from areal data of the same attribute is addressed within the general geostatistical framework of change of support (the term support referring to the domain informed by each measurement). It is shown that the geostatistical framework: (i) can explicitly and consistently account for the support difference between the available areal data and the sought-after point predictions, (ii) yields coherent (mass-preserving or pycnophylactic) predictions, and (iii) provides a measure of reliability (standard error) associated with each prediction. In the case of stochastic simulation, the resulting simulated realizations reproduce a point-support histogram (Gaussian in this work) and a semivariogram model (possibly including anisotropic nested structures).

A case study using simulated areal data illustrates the application of the proposed methodology in a remote sensing context, whereby the areal data are available at a regular pixel support. It is demonstrated that point-support (sub-pixel scale) predictions and simulated realizations can be readily obtained, and that such predictions/simulations are consistent with the available information at the coarser (pixel-level) resolution.

The point-support predictions/simulations obtained by the proposed methodology can be used to assess the uncertainty in spatially distributed environmental model outputs due to uncertain input parameters linked to coarser resolution data.

1. Introduction

In many scientific disciplines, spatially distributed models simulating physical or human-related processes require input parameter maps at a finer resolution than what is often available. In hydrometeorology, for example, detailed modeling calls for downscaling predictions of climate-related variables produced by global circulation models (GCMs) to finer resolutions for climate change impact assessment studies. Along the same lines, remotely sensed information must be often processed to obtain finer resolution inputs for spatially explicit ecological models. For a recent review of such scale issues in geography and some geostatistical solutions, the reader is referred to Atkinson and Tate (2000).

The transformation of coarse resolution data to finer resolution maps is often termed downscaling, and it is a particular case of change of support (the term support referring to the domain informed by each measurement), see for example Gotway and Young (2002). In this paper, we focus on a particular case of downscaling, that of predicting point-support values from areal data, a procedure also termed area-to-point interpolation. Existing methods for area-to-point interpolation in geography, see for example Lam (1983), tend to ignore one or more of the following critical issues: (i) the explicit account of the different data supports, (ii) the coherence of predictions: the areal-average, for example, of point predictions within any areal support should be equal to the corresponding areal datum (if the latter is assumed error-free), and (iii) the assessment of uncertainty regarding the resulting point predictions. Perhaps the only existing (albeit very specific) method of area-to-point interpolation that satisfies requirements (i) and (ii), but not (iii), is the pycnophylactic interpolation method of Tobler (1979).

In this work, we generalize the geostatistical framework proposed by Kyriakidis (2003) for solving the area-to-point spatial interpolation problem, under the constraint of meeting all the above three requirements. More precisely, we account for areal data defined as convolutions of point-support values with an appropriate sampling kernel, which includes summation and averaging as special cases. We extend the above framework to the case of stochastic simulation, i.e., to the generation of alternative simulated realizations of point-support values conditioned to the available areal data. We prove that the area-to-point predictions and simulations are coherent, i.e., that these point-support predictions/simulations, when convolved with the sampling kernel, reproduce exactly the corresponding areal data. As an aside, we also demonstrate that the choropleth map solution can be derived as a particular case of the geostatistical approach. Last, we illustrate (for the multivariate Gaussian case) that the alternative simulated point-support realizations reproduce, in addition to the areal data, a point-support Gaussian histogram and a point-support semivariogram. Such point-support simulated realizations can be used in a Monte Carlo framework to assess the uncertainty in model outputs due to uncertain model inputs (subject to data constraints at a coarser resolution). A case study based on a simulated set of areal data is used to illustrate the application of the proposed methodology in a remote sensing context.

2. Theory

Consider the task of point prediction at a set of P prediction locations $\{\mathbf{u}_p, p = 1, \ldots, P\}$ within a study area A, where $\mathbf{u}_p$ denotes the coordinate vector of the p-th location. The available areal data are defined on a set of K supports $\{v_k, k = 1, \ldots, K\}$, where v_k denotes the k-th support with centroid $\mathbf{u}_k$ and otherwise arbitrary shape. Typically, the number K of areal supports is much smaller than the number P of prediction locations, i.e., $K << P$. In this paper, the measure (length in 1D, area in 2D, volume in 3D) of each support is assumed constant, and denoted as $|v|$. For simplicity, we assume that each support v_k contains the same number of P_k prediction locations: $P_k = P/K, \forall k$, with $\sum_{k=1}^{K} P_k = P$. An example of such a setting is that of remotely sensed imagery, whereby the reported reflectance values constitute integrated measurements over a constant pixel support. The objective in this case would be that of point prediction at a sub-pixel level from the available pixel-support data. The case of unequal areal data supports is addressed in Kyriakidis

(2003) and not considered in this paper.

2.1 Spatial characteristics of point values and areal data

Let $z(\mathbf{u}_p)$ denote the true (but unknown) point-support value at location $\mathbf{u}_p$. In the geostatistical framework, that unknown value $z(\mathbf{u}_p)$ is viewed as a realization of a random variable (RV) $Z(\mathbf{u}_p)$. The set of all point RVs in the study domain $\{Z(\mathbf{u}_p), p = 1,\ldots,P\}$ constitutes a discrete random function (RF), which is partially characterized by its first two moments (mean and covariance).

In this paper, we focus on the second-order stationary case, whereby:

- the expected value (mean) of any RV $Z(\mathbf{u}_p)$ is constant within A:

$$E\{Z(\mathbf{u}_p)\} = m_Z, \ \ \forall \mathbf{u}_p \in A \tag{1}$$

- the covariance between any two RVs $Z(\mathbf{u}_{p'})$ and $Z(\mathbf{u}_p)$ at locations $\mathbf{u}_{p'} = \mathbf{u}_p + \mathbf{h}$ and $\mathbf{u}_p$ separated by a vector $\mathbf{h}$, depends only on (the modulus and possibly orientation of) that vector $\mathbf{h}$:

$$Cov\{Z(\mathbf{u}_{p'}), Z(\mathbf{u}_p)\} = E\{[Z(\mathbf{u}_p+\mathbf{h}) - m_Z][Z(\mathbf{u}_p) - m_Z]\} = C_Z(\mathbf{h}), \ \ \forall \mathbf{u}_{p'}, \mathbf{u}_p, \mathbf{h} \tag{2}$$

- in the case of a finite variance $Var\{Z(\mathbf{u}_p)\} = C_Z(\mathbf{0})$, the semivariogram of the RF $\{Z(\mathbf{u}_p), p = 1,\ldots,P\}$ is obtained as:

$$\gamma_Z(\mathbf{h}) = \frac{1}{2}E\{[Z(\mathbf{u}_p+\mathbf{h}) - Z(\mathbf{u}_p)]^2\} = C_Z(\mathbf{0}) - C_Z(\mathbf{h}), \ \ \forall \mathbf{u}_{p'}, \mathbf{u}_p, \mathbf{h} \tag{3}$$

  Note that in the case of an infinite variance, $Var\{Z(\mathbf{u}_p)\} \to \infty$, the semivariogram $\gamma_Z(\mathbf{h})$ is still defined, even if the covariance $C_Z(\mathbf{h})$ is not.

The areal datum corresponding to the k-th support v_k is denoted as $d(v_k)$, and is linked to the point-support values via a sampling function g_k:

$$d(v_k) = \sum_{p=1}^{P} g_k(\mathbf{u}_p) z(\mathbf{u}_p) \tag{4}$$

where $g_k(\mathbf{u}_p)$ denotes the value of the sampling function with respect to the k-th support v_k at location $\mathbf{u}_p$. The above definition constitutes a convolution of the point-support values with the sampling function g_k; this latter function plays the role of a convolution kernel.

Some typical examples of that sampling function include:

1. an indicator function $g_k(\mathbf{u}_p) = 1$, if $\mathbf{u}_p \in v_k$, zero if not. In this case, the areal datum $d(v_k)$ is simply the sum of all P_k point values within support v_k.

2. the indicator function normalized by the total number P_k of points within any support v_k: $g_k(\mathbf{u}_p) = 1/P_k$ if $\mathbf{u}_p \in v_k$, zero if not. In this case, the areal datum $d(v_k)$ is simply the mean of all P_k point values within support v_k.

3. a more elaborate kernel function, such as the point-spread function of a satellite, weighting the contribution of each point value $z(\mathbf{u}_p)$ within v_k to the areal datum $d(v_k)$ (Vanmarcke, 1983; Collins and Woodcock, 1999).

In the geostatistical framework, the k-th areal datum $d(v_k)$ is viewed as a realization of a new RV $D(v_k)$. The set of all K such areal RVs $\{D(v_k), k = 1, \ldots, K\}$ constitutes a new discrete random function, whose moments are functionally related to those of the point RF $\{Z(\mathbf{u}_p), p = 1, \ldots, P\}$ (Anderson, 1958; Matheron, 1971; Journel and Huijbregts, 1978). More precisely:

- the expected value of any areal datum RV $D(v_k)$ is constant within A, and equal to:

$$
\begin{aligned}
E\{D(v_k)\} &= E\left\{\sum_{p=1}^{P} g_k(\mathbf{u}_p) Z(\mathbf{u}_p)\right\} = \sum_{p=1}^{P} g_k(\mathbf{u}_p) E\left\{Z(\mathbf{u}_p)\right\} \qquad (5)\\
&= m_Z \sum_{p=1}^{P} g_k(\mathbf{u}_p) = m_D, \ \ \forall v_k \in A
\end{aligned}
$$

  or, in words, the mean m_D of the areal-support data is a linear function of the point-support mean m_Z. Note that if the sum of the sampling function evaluated over v_k is equal to one, i.e., if $\sum_{p=1}^{P} g_k(\mathbf{u}_p) = 1$, the mean of the areal-support RF $\{D(v_k), k = 1, \ldots, K\}$ is equal to that of the point-support RF $\{Z(\mathbf{u}_p), p = 1, \ldots, P\}$, i.e., $m_D = m_Z$; this is the case of an areal datum $d(v_k)$ defined as the mean of P_k point values within support v_k.

- the covariance between any two areal data RVs $D(v_{k'})$ and $D(v_k)$ is obtained as:

$$
\begin{aligned}
Cov\{D(v_{k'}), D(v_k)\} &= Cov\left\{\left[\sum_{p'=1}^{P} g_{k'}(\mathbf{u}_{p'}) Z(\mathbf{u}_{p'})\right], \left[\sum_{p=1}^{P} g_k(\mathbf{u}_p) Z(\mathbf{u}_p)\right]\right\} \qquad (6)\\
&= \sum_{p'=1}^{P} g_{k'}(\mathbf{u}_{p'}) \sum_{p=1}^{P} g_k(\mathbf{u}_p) Cov\{Z(\mathbf{u}_{p'}) Z(\mathbf{u}_p)\}\\
&= \sum_{p'=1}^{P} \sum_{p=1}^{P} g_{k'}(\mathbf{u}_{p'}) g_k(\mathbf{u}_p) C_Z(\mathbf{u}_{p'} - \mathbf{u}_p) = C_D(v_{k'}, v_k)
\end{aligned}
$$

  or, in words, the covariance $C_D(v_{k'}, v_k)$ between any two supports $v_{k'}$ and v_k is a double weighted linear combination of point covariance values $C_Z(\mathbf{u}_{p'} - \mathbf{u}_p)$ corresponding to all possible vectors formed by any two prediction locations $\mathbf{u}_{p'} \in v_{k'}$ and $\mathbf{u}_p \in v_k$. The semivariogram $\gamma_D(v_{k'}, v_k)$ between any two supports $v_{k'}$ and v_k can be derived in a similar way from the point semivariogram $\gamma_Z(\mathbf{h})$.

The link between the point-support RF $\{Z(\mathbf{u}_p), p = 1, \ldots, P\}$ and the areal-support RF $\{D(v_k), k = 1, \ldots, K\}$ is the cross-covariance between any point RV $Z(\mathbf{u}_{p'})$ and any areal datum RV $D(v_k)$:

$$
\begin{aligned}
Cov\{Z(\mathbf{u}_{p'}), D(v_k)\} &= Cov\left\{Z(\mathbf{u}_{p'}), \left[\sum_{p=1}^{P} g_k(\mathbf{u}_p) Z(\mathbf{u}_p)\right]\right\} \qquad (7)\\
&= \sum_{p=1}^{P} g_k(\mathbf{u}_p) Cov\{Z(\mathbf{u}_{p'}) Z(\mathbf{u}_p)\}
\end{aligned}
$$

$$= \sum_{p=1}^{P} g_k(\mathbf{u}_p) C_Z(\mathbf{u}_{p'} - \mathbf{u}_p) = C_{ZD}(\mathbf{u}_{p'}, v_k)$$

In what follows, we provide the vector/matrix expressions of the point-support and areal-support spatial characteristics; these expressions are used extensively hereafter. Vectors are denoted with lower case boldface letters, while matrices are denoted with upper case boldface letters.

Let $\mathbf{z} = [z(\mathbf{u}_p), p = 1, \ldots, P]'$ denote the $(P \times 1)$ vector of unknown point-support values, and $\mathbf{d} = [d(v_k), k = 1, \ldots, K]'$ denote the $(K \times 1)$ vector of areal-support data; here superscript $'$ denotes transposition. The two vectors $\mathbf{z}$ and $\mathbf{d}$ are linked as:

$$\mathbf{d} = \mathbf{G}\mathbf{z} \tag{8}$$

where $\mathbf{G}$ is a $(K \times P)$ matrix, whose k-th row consists of the $(1 \times P)$ vector of sampling function values for support v_k: $\mathbf{g}_k = [g_k(\mathbf{u}_p), p = 1, \ldots, P]$.

Let $\mathbf{m}_Z = [m_Z(\mathbf{u}_p), p = 1, \ldots, P]'$ denote the $(P \times 1)$ vector of constant mean values at the P prediction locations. The $(K \times 1)$ vector of constant mean values of the K areal data RVs is denoted as $\mathbf{m}_D = [m_D(v_k), k = 1, \ldots, K]'$, and is related to $\mathbf{m}_Z$ as:

$$\mathbf{m}_D = E\{\mathbf{d}\} = E\{\mathbf{G}\mathbf{z}\} = \mathbf{G}E\{\mathbf{z}\} = \mathbf{G}\mathbf{m}_Z \tag{9}$$

The $(P \times P)$ matrix $\mathbf{C}_Z = [C_Z(\mathbf{u}_{p'} - \mathbf{u}_p), p', p = 1, \ldots, P]$ of point covariance values between all pairs of prediction locations is defined as:

$$\mathbf{C}_Z = E\{[\mathbf{z} - \mathbf{m}_Z][\mathbf{z} - \mathbf{m}_Z]'\} \tag{10}$$

where $\mathbf{C}_Z$ is positive definite, if its entries have been computed using a positive definite covariance model, and symmetric, which entails $\mathbf{C}_Z = \mathbf{C}_Z'$.

The $(P \times K)$ matrix $\mathbf{C}_{ZD} = [C_{ZD}(\mathbf{u}_{p'}, v_k), p' = 1, \ldots, P, k = 1, \ldots, K]$ of cross-covariance values between all pairs of prediction locations and areal data is defined as:

$$\begin{aligned} \mathbf{C}_{ZD} &= E\{[\mathbf{z} - \mathbf{m}_Z][\mathbf{d} - \mathbf{m}_D]'\} = E\{[\mathbf{z} - \mathbf{m}_Z][\mathbf{G}\mathbf{z} - \mathbf{G}\mathbf{m}_Z]'\} \\ &= E\{[\mathbf{z} - \mathbf{m}_Z][\mathbf{z} - \mathbf{m}_Z]'\}\mathbf{G}' = \mathbf{C}_Z\mathbf{G}' \end{aligned} \tag{11}$$

Note that, per symmetry, $\mathbf{C}_{ZD}' = (\mathbf{C}_Z\mathbf{G}')' = \mathbf{G}\mathbf{C}_Z = \mathbf{C}_{DZ}$.

The $(K \times K)$ matrix $\mathbf{C}_D = [C_D(v_{k'}, v_k), k', k = 1, \ldots, K]$ of covariance values between all pairs of areal data is defined as:

$$\begin{aligned} \mathbf{C}_D &= E\{[\mathbf{d} - \mathbf{m}_D][\mathbf{d} - \mathbf{m}_D]'\} = E\{[\mathbf{G}\mathbf{z} - \mathbf{G}\mathbf{m}_Z][\mathbf{G}\mathbf{z} - \mathbf{G}\mathbf{m}_Z]'\} \\ &= E\{\mathbf{G}[\mathbf{z} - \mathbf{m}_Z][\mathbf{z} - \mathbf{m}_Z]'\mathbf{G}'\} = \mathbf{G}E\{[\mathbf{z} - \mathbf{m}_Z][\mathbf{z} - \mathbf{m}_Z]'\}\mathbf{G}' \\ &= \mathbf{G}\mathbf{C}_Z\mathbf{G}' = \mathbf{G}\mathbf{C}_{ZD} \end{aligned} \tag{12}$$

In this paper, we focus on the multivariate Gaussian case, whereby the point values are distributed as: $\mathbf{z} \sim N(\mathbf{m}_Z, \mathbf{C}_Z)$. Since the area-support data are defined as linear combinations of point-support values, their distribution is also multivariate Gaussian (Anderson, 1958), i.e.: $\mathbf{d} \sim N(\mathbf{m}_D, \mathbf{C}_D) = N(\mathbf{G}\mathbf{m}_Z, \mathbf{G}\mathbf{C}_Z\mathbf{G}')$.

In practice, it is only the data vector $\mathbf{d}$ that is observable, and consequently only $\mathbf{m}_D$ and $\mathbf{C}_D$ can be estimated directly. The computation of $\mathbf{m}_Z$ and $\mathbf{C}_Z$ constitutes an ill-posed inverse problem, see for example Menke (1989), which can be solved using deconvolution techniques. In this paper, we do not address the deconvolution of the area-support covariance matrix $\mathbf{C}_D$ to infer the point-support covariance model $C_Z(\mathbf{h})$, and instead we assume that $C_Z(\mathbf{h})$ is known. The reader is referred to Journel and Huijbregts (1978) or Collins and Woodcock (1999) for details regarding the estimation of $C_Z(\mathbf{h})$ from areal data.

2.2 Area-to-point kriging

Consider now the task of predicting the unknown point value $z(\mathbf{u}_p)$ at the p-th location $\mathbf{u}_p$. The area-to-point simple kriging prediction $\hat{z}(\mathbf{u}_p)$ is expressed as a weighted linear combination of K areal data comprising the $(K \times 1)$ data vector $\mathbf{d}$:

$$\hat{z}(\mathbf{u}_p) = m_Z + \mathbf{w}'_p[\mathbf{d} - \mathbf{m}_D] = m_Z + \sum_{k=1}^{K} w_k(\mathbf{u}_p)[d(v_k) - m_D] \tag{13}$$

where $\mathbf{w}_p = [w_k(\mathbf{u}_p), k = 1, \ldots, K]'$ denotes a $(K \times 1)$ vector of weights, whose k-th entry $w_k(\mathbf{u}_p)$ is the weight applied to the k-th areal datum $d(v_k)$ for prediction at the p-th location $\mathbf{u}_p$. The constant mean value m_Z is assumed known, or can be estimated by inverting Equation (5) as: $m_Z = m_D / \sum_{p=1}^{P} g_k(\mathbf{u}_p)$.

The vector of kriging weights $\mathbf{w}_p$ for prediction location $\mathbf{u}_p$ is derived per solution of the system of normal equations, see Kyriakidis (2003):

$$\mathbf{C}_D \mathbf{w}_p = \mathbf{c}^p_{ZD} \tag{14}$$

where $\mathbf{c}^p_{ZD} = [C_{ZD}(\mathbf{u}_p, v_k), k = 1, \ldots, K]'$ denotes the $(K \times 1)$ vector of cross-covariance values between the p-th prediction location and all K areal data supports.

The vector $\mathbf{w}_p$ is then computed as:

$$\mathbf{w}_p = \mathbf{C}_D^{-1}\mathbf{c}^p_{ZD} \Rightarrow \mathbf{w}'_p = (\mathbf{c}^p_{ZD})'\mathbf{C}_D^{-1} \tag{15}$$

where $\mathbf{C}_D^{-1}$ denotes the inverse of the areal-support covariance matrix $\mathbf{C}_D$. Note that, due to $\mathbf{C}_D$ being symmetric, $(\mathbf{C}_D^{-1})' = \mathbf{C}_D^{-1}$.

The prediction error variance $\hat{\sigma}^2(\mathbf{u}_p)$ at the p-th location $\mathbf{u}_p$ is written as:

$$\hat{\sigma}^2(\mathbf{u}_p) = C_Z(\mathbf{0}) - \mathbf{w}'_p\mathbf{c}^p_{ZD} = C_Z(\mathbf{0}) - (\mathbf{c}^p_{ZD})'\mathbf{C}_D^{-1}\mathbf{c}^p_{ZD} \tag{16}$$

and is homoscedastic, i.e., does not depend on the areal data values but only on the configuration and shape of their supports, as well as the point covariance model $C_Z(\mathbf{h})$.

Combining Equations (13) and (15), the $(P \times 1)$ vector $\hat{\mathbf{z}} = [\hat{z}(\mathbf{u}_p), p = 1, \ldots, P]'$ of all P area-to-point kriging predictions at all P locations is written as:

$$\hat{\mathbf{z}} = \mathbf{m}_Z + \mathbf{W}[\mathbf{d} - \mathbf{m}_D] = \mathbf{m}_Z + \mathbf{C}_{ZD}\mathbf{C}_D^{-1}[\mathbf{d} - \mathbf{m}_D] \tag{17}$$

where $\mathbf{W} = [w_k(\mathbf{u}_p), p = 1, \ldots, P, k = 1, \ldots, K]$ is a $(P \times K)$ matrix of weights whose p-th row is the vector $\mathbf{w}'_p$, and $\mathbf{C}_{ZD}$ is the $(P \times K)$ matrix of cross-covariances computed from Equation (11) whose p-th row is the vector $(\mathbf{c}^p_{ZD})'$.

The area-to-point predictions are unbiased, i.e., the expected value of vector $\hat{\mathbf{z}}$ is equal to the mean vector $\mathbf{m}_Z$. Indeed, from Equation (17):

$$\mathbf{m}_{\hat{Z}} = E\{\hat{\mathbf{z}}\} = E\{\mathbf{m}_Z\} + \mathbf{W}E\{[\mathbf{d} - \mathbf{m}_D]\} = \mathbf{m}_Z + \mathbf{W}[\mathbf{m}_D - \mathbf{m}_D] = \mathbf{m}_Z \tag{18}$$

which also entails that the prediction error has a zero expectation, i.e., $\mathbf{m}_{Z-\hat{Z}} = E\{\mathbf{z} - \hat{\mathbf{z}}\} = 0$.

It is also straightforward to prove that the resulting area-to-point kriging predictions reproduce the areal-support data, i.e., to prove that $\mathbf{G}\hat{\mathbf{z}} = \mathbf{d}$. Indeed, using Equation (17):

$$\mathbf{G}\hat{\mathbf{z}} = \mathbf{G}\left(\mathbf{m}_Z + \mathbf{C}_{ZD}\mathbf{C}_D^{-1}[\mathbf{d} - \mathbf{m}_D]\right) = \mathbf{G}\mathbf{m}_Z + \mathbf{G}\mathbf{C}_{ZD}\mathbf{C}_D^{-1}[\mathbf{d} - \mathbf{m}_D] \tag{19}$$

Using Equation (9) and the last relation of Equation (12), the above equation becomes:

$$\mathbf{G}\hat{\mathbf{z}} = \mathbf{m}_D + \mathbf{C}_D\mathbf{C}_D^{-1}[\mathbf{d} - \mathbf{m}_D] = \mathbf{m}_D + \mathbf{d} - \mathbf{m}_D = \mathbf{d} \tag{20}$$

which entails that any areal-support datum $d(v_k)$ can be reproduced exactly by the resulting area-to-point predictions, as long as the latter are computed by taking into account the functional relationship between that areal datum and the point data via Equations (8) through (17). It should be stressed here that this coherence characteristic of the area-to-point kriging predictions: (i) does not call for any multivariate Gaussian assumption regarding the vectors $\mathbf{d}$ or $\mathbf{z}$, and (ii) is independent of the particular point covariance model $C_Z(\mathbf{h})$ used for prediction.

An extremely particular case

Assume, for a moment, that the mean vectors $\mathbf{m}_Z$ and $\mathbf{m}_D$ are zero vectors, and that the point-support values are spatially uncorrelated, i.e., $\mathbf{C}_Z^u = \mathbf{I}$, where $\mathbf{I}$ denotes the identity matrix (here of dimensions $(P \times P)$), and superscript u emphasizes the assumption of uncorrelated point values. It readily follows from Equations (11) and (12) that: $\mathbf{C}_{ZD}^u = \mathbf{G}'$, and $\mathbf{C}_D^u = \mathbf{G}\mathbf{G}'$. Consequently, Equation (17) becomes:

$$\hat{\mathbf{z}}^u = \mathbf{G}'(\mathbf{G}\mathbf{G}')^{-1}\mathbf{d} \tag{21}$$

which is also termed the minimum length solution of an under-determined inverse problem (Menke, 1989).

We will show, in terms of a small example, that this solution is the same as the choropleth map solution frequently used in mapping, whereby the areal datum is equally distributed to all points within a support. Consider, for example, the case of two areal data, i.e., $\mathbf{d} = [4 \;\; 6]'$, each pertaining to the sum of two point-support values within a 1D segment. In this case, Equation (21) yields:

$$\hat{\mathbf{z}}^u = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \\ 3 \end{bmatrix} \tag{22}$$

which is precisely the choropleth map solution.

A non-diagonal point-support covariance matrix $\mathbf{C}_Z \neq \mathbf{I}$ can be therefore regarded as a filter that modifies the minimum length solution depending on the type, range, and nested structures of the point-support covariance model $C_Z(\mathbf{h})$. This notion of filtering is termed regularization in inverse problem theory, and should not be confused with the geostatistical concept of regularization (the latter being a particular case of convolution).

Going back to the general case of an arbitrary (yet positive definite) point-support covariance matrix $\mathbf{C}_Z$, the $(P \times P)$ matrix $\mathbf{C}_{\hat{Z}}$ of covariance values between all possible pairs of point predictions is derived as:

$$\begin{aligned}
\mathbf{C}_{\hat{Z}} &= E\{[\hat{\mathbf{z}} - \mathbf{m}_{\hat{Z}}][\hat{\mathbf{z}} - \mathbf{m}_{\hat{Z}}]'\} = E\{\hat{\mathbf{z}}\hat{\mathbf{z}}'\} - \mathbf{m}_Z \mathbf{m}_Z' \\
&= E\{(\mathbf{m}_Z + \mathbf{C}_{ZD}\mathbf{C}_D^{-1}[\mathbf{d} - \mathbf{m}_D])(\mathbf{m}_Z + \mathbf{C}_{ZD}\mathbf{C}_D^{-1}[\mathbf{d} - \mathbf{m}_D])'\} - \mathbf{m}_Z \mathbf{m}_Z' \\
&= \mathbf{C}_{ZD}\mathbf{C}_D^{-1} E\{[\mathbf{d} - \mathbf{m}_D][\mathbf{d} - \mathbf{m}_D]'\}\mathbf{C}_D^{-1}\mathbf{C}_{DZ} = \mathbf{C}_{ZD}\mathbf{C}_D^{-1}\mathbf{C}_{DZ}
\end{aligned} \tag{23}$$

The $(P \times P)$ matrix $\mathbf{C}_{Z\hat{Z}}$ of cross-covariance values between all possible pairs of true point-support values and point predictions is written as:

$$\begin{aligned}
\mathbf{C}_{Z\hat{Z}} &= E\{[\mathbf{z} - \mathbf{m}_Z][\hat{\mathbf{z}} - \mathbf{m}_{\hat{Z}}]'\} = E\{\mathbf{z}\hat{\mathbf{z}}'\} - E\{\mathbf{m}_Z \mathbf{m}_{\hat{Z}}'\} \\
&= E\{\mathbf{z}(\mathbf{m}_Z + \mathbf{C}_{ZD}\mathbf{C}_D^{-1}[\mathbf{d} - \mathbf{m}_D])'\} - \mathbf{m}_Z \mathbf{m}_Z' \\
&= E\{\mathbf{z}\}\mathbf{m}_Z' + E\{\mathbf{z}[\mathbf{d} - \mathbf{m}_D]'\}\mathbf{C}_D^{-1}\mathbf{C}_{DZ} - \mathbf{m}_Z \mathbf{m}_Z' = \mathbf{C}_{ZD}\mathbf{C}_D^{-1}\mathbf{C}_{DZ} = \mathbf{C}_{\hat{Z}Z}
\end{aligned} \tag{24}$$

Per the bi-linearity property of the covariance operator, the $(P \times P)$ matrix $\mathbf{C}_{Z-\hat{Z}}$ of covariance values between all pairs of prediction errors is written as:

$$\mathbf{C}_{Z-\hat{Z}} = \mathbf{C}_Z - \mathbf{C}_{Z\hat{Z}} - \mathbf{C}_{\hat{Z}Z} + \mathbf{C}_{\hat{Z}} = \mathbf{C}_Z - \mathbf{C}_{Z\hat{Z}} = \mathbf{C}_Z - \mathbf{C}_{ZD}\mathbf{C}_D^{-1}\mathbf{C}_{DZ} \tag{25}$$

using the fact that $\mathbf{C}_{\hat{Z}} = \mathbf{C}_{Z\hat{Z}} = \mathbf{C}_{\hat{Z}Z}$ from Equations (23) and (24). The $(P \times 1)$ vector of diagonal entries, $diag(\mathbf{C}_{Z-\hat{Z}})$, constitutes the $(P \times 1)$ vector of prediction error variances at the P prediction locations. The p-th entry of that vector is $\hat{\sigma}^2(\mathbf{u}_p)$ given in Equation (16).

Last, the point predictions are uncorrelated with the prediction errors, i.e., the $(P \times P)$ matrix $\mathbf{C}_{\hat{Z},Z-\hat{Z}}$ of covariance values between all pairs of point predictions and prediction errors is the zero matrix:

$$\mathbf{C}_{\hat{Z},Z-\hat{Z}} = \mathbf{C}_{\hat{Z}Z} - \mathbf{C}_{\hat{Z}\hat{Z}} = \mathbf{0} \tag{26}$$

using, again, Equations (23) and (24).

It should be stressed that all the above covariance matrices are homoscedastic, i.e., they do not depend on the areal data values. They only depend on: (i) the spatial configuration and shape of the areal supports, (ii) the locus of the prediction locations within each support, and (iii) the point-support covariance model $C_Z(\mathbf{h})$.

The $(P \times 1)$ vector $\mathbf{z}$ of true point-support values can always be decomposed into:

$$\mathbf{z} = \hat{\mathbf{z}} + [\mathbf{z} - \hat{\mathbf{z}}] = \hat{\mathbf{z}} + \mathbf{e} \tag{27}$$

where $\mathbf{e} = [\mathbf{z} - \hat{\mathbf{z}}]$ denotes a $(P \times 1)$ vector of prediction errors. Since, per Equation (18), $\mathbf{m}_{\hat{Z}} = \mathbf{m}_Z$, it follows that the prediction error component $\mathbf{e}$ has zero mean. In addition, per Equation (26), that error component $\mathbf{e}$ is uncorrelated with (and in the multivariate Gaussian case independent of) the kriging component $\hat{\mathbf{z}}$.

From the above, it follows that the $(P \times P)$ covariance matrix $\mathbf{C}_Z$ of the point-support values can also be decomposed as:

$$\mathbf{C}_Z = \mathbf{C}_{\hat{Z}} + \mathbf{C}_{Z-\hat{Z}} \tag{28}$$

which can be trivially proved from Equations (23) and (25).

It can therefore be deduced that the covariance matrix $\mathbf{C}_{\hat{Z}}$ of area-to-point predictions is not the same as the covariance matrix $\mathbf{C}_Z$ of the true point-support values; that covariance deficit is none other than the covariance matrix $\mathbf{C}_{Z-\hat{Z}}$ of prediction errors. In many applications, where z-maps are used as inputs to non-linear spatially distributed models, however, it is of utmost importance that such model inputs have the correct covariance in order to be able to realistically assess the uncertainty in model outputs.

One way of generating a realization of point-support values with covariance matrix $\mathbf{C}_Z$ is to add to the area-to-point predictions a simulated error component with covariance matrix $\mathbf{C}_{Z-\hat{Z}}$. This covariance matrix, however, is non-stationary: its diagonal elements, for example, are not equal. In what follows, we illustrate precisely the construction of point-support realizations of prediction error with that non-stationary covariance matrix $\mathbf{C}_{Z-\hat{Z}}$, which, when added to the kriging predictions $\hat{\mathbf{z}}$, yield realizations with covariance matrix $\mathbf{C}_Z$ (over a large number of such simulated realizations).

2.3 Area-to-point stochastic simulation

A straightforward method for generating a simulated realization with an arbitrary (yet positive definite) non-stationary covariance is simulation via Cholesky decomposition, see for example Deutsch and Journel (1998). This method, however, cannot handle large grids since one has to compute the Cholesky decomposition of a, typically very large, $(P \times P)$ covariance matrix $\mathbf{C}_{Z-\hat{Z}}$.

An alternative simulation method, which does not suffer from the above limitation, is that of kriging error simulation (Journel and Huijbregts, 1978; Deutsch and Journel, 1998; Chilès and Delfiner, 1999). In a nutshell, kriging error simulation aims at mimicking (simulating) the prediction error incurred by kriging, by repeating the kriging procedure using a simulated data set with the same configuration as the original data. The simulated data are extracted from an unconditional realization with covariance model $C_Z(\mathbf{h})$. Since both the simulated kriging and the unconditional realization are available, one can readily compute a simulated prediction error realization. In what follows, we extend the above method to condition point-support realizations with covariance matrix $\mathbf{C}_Z$ to the areal-support data vector $\mathbf{d}$.

Let $\mathbf{z}^{(s)}$ denote a $(P \times 1)$ vector of simulated point-support values with stationary mean vector $\mathbf{m}_Z^{(s)} \simeq \mathbf{m}_Z$ and covariance matrix $\mathbf{C}_Z^{(s)} \simeq \mathbf{C}_Z$. Here superscript (s) denotes the s-th realization, and there might be S such simulated vectors $\{\mathbf{z}^{(s)}, s = 1, \ldots, S\}$. These S unconditional point-support realizations can be generated by various stochastic simulation algorithms, see for example Deutsch and Journel (1998). An extremely fast algorithm that can also handle large grids is moving average simulation in the frequency domain (Oliver, 1995; Le Ravalec et al., 2000).

The key concept of moving average simulation is that a realization of a Gaussian white noise process (with complete absence of spatial correlation) can be transformed (smoothed) to a realization with a desired covariance model $C_Z(\mathbf{h})$ via convolution by an appropriate kernel. The entries of that kernel can be directly derived from the covariance model $C_Z(\mathbf{h})$, and the convolution can be performed much faster in the frequency domain by means of the Fast Fourier Transform (FFT).

The flowchart of frequency domain, moving average simulation (on a regular grid) of a Gaussian process with mean m_Z and point-support covariance model $C_Z(\mathbf{h})$ is given below (for more details, see Le Ravalec et al. (2000)):

1. Construction of a covariance map on the simulation grid using the covariance model $C_Z(\mathbf{h})$, and determination of the associated power spectrum by computing the FFT of that covariance map.

2. Generation of a standard Gaussian realization (with white noise covariance) on the same grid, and computation of its FFT.

3. Element by element multiplication of the square root of the power spectrum and the result of the previous step. This product constitutes a simulated realization in the frequency domain.

4. Inverse FFT of the above product to obtain a simulated realization in the original data space. A constant value m_Z is added to the result to obtain a realization $\mathbf{z}^{(s)}$ with mean m_Z.

A new realization $\mathbf{z}^{(s')}$ can be generated by repeating steps 2-4 with a new white noise image simulated in step 2.

By applying Equation (8), i.e., by convolving the simulated point values within each support v_k with the convolution kernel g_k, one can generate a $(K \times 1)$ vector $\mathbf{d}^{(s)}$ of simulated areal-support data with the *same* support configuration as the original data $\mathbf{d}$:

$$\mathbf{d}^{(s)} = \mathbf{G}\mathbf{z}^{(s)} \tag{29}$$

where, per Equation (9), the mean vector of simulated areal data is $\mathbf{m}_D^{(s)} \simeq \mathbf{G}\mathbf{m}_Z$, and per Equation (12), the $(K \times K)$ covariance matrix of the simulated areal data is: $\mathbf{C}_D^{(s)} \simeq \mathbf{G}\mathbf{C}_{ZD} = \mathbf{G}\mathbf{C}_Z\mathbf{G}'$.

Consider now generating a new simulated realization, i.e., a new $(K \times 1)$ vector $\mathbf{z}_c^{(s)}$ of simulated values, conditioned (hence the subscript c) on the areal-support data vector $\mathbf{d}$ as:

$$\mathbf{z}_c^{(s)} = \hat{\mathbf{z}} + \mathbf{e}^{(s)} = \hat{\mathbf{z}} + [\mathbf{z}^{(s)} - \hat{\mathbf{z}}^{(s)}] = \hat{\mathbf{z}} + [\mathbf{z}^{(s)} - \mathbf{C}_{ZD}\mathbf{C}_D^{-1}\mathbf{d}^{(s)}] \tag{30}$$

where $\hat{\mathbf{z}}^{(s)}$ is a $(P \times 1)$ vector of simple area-to-point kriging predictions obtained from the simulated areal-support data $\mathbf{d}^{(s)}$ of Equation (29), and $\mathbf{e}^{(s)} = [\mathbf{z}^{(s)} - \hat{\mathbf{z}}^{(s)}]$ is a $(P \times 1)$ vector of simulated prediction error values.

The mean of the conditionally simulated point values is approximately equal to $\mathbf{m}_Z$:

$$E\{\mathbf{z}_c^{(s)}\} = E\{\hat{\mathbf{z}}\} + [E\{\mathbf{z}^{(s)} - \hat{\mathbf{z}}^{(s)}\}] \simeq \mathbf{m}_Z + [\mathbf{m}_Z - \mathbf{m}_Z] = \mathbf{m}_Z \tag{31}$$

since, per Equation (18), $E\{\hat{\mathbf{z}}^{(s)}\} = \mathbf{m}_Z^{(s)} \simeq \mathbf{m}_Z$. In words, because the area-to-point predictions are unbiased, area-to-point kriging using simulated areal data with mean $\simeq \mathbf{m}_D$ ensures that the simulated point predictions have mean $\simeq \mathbf{m}_Z$.

Recall that: (i) the vector $\hat{\mathbf{z}}^{(s)}$ of simulated area-to-point kriging predictions is derived from simulated areal data $\mathbf{d}^{(s)}$ that have the same support configuration as the original areal data $\mathbf{d}$, and (ii) the vector $\mathbf{z}^{(s)}$ of point-support simulated values has (approximately) covariance matrix $\mathbf{C}_Z$. This entails that the simulated kriging component $\hat{\mathbf{z}}^{(s)}$ has the same covariance matrix with that of Equation (23), i.e.: $\mathbf{C}_{\hat{Z}^{(s)}} \simeq \mathbf{C}_{\hat{Z}}$, since the latter depends only on the areal data configuration and on the point covariance model $C_Z(\mathbf{h})$. Consequently, the resulting simulated prediction error component $\mathbf{e}^{(s)} = [\mathbf{z}^{(s)} - \hat{\mathbf{z}}^{(s)}]$: (i) has the covariance matrix of Equation (25), i.e.: $\mathbf{C}_{Z^{(s)}-\hat{Z}^{(s)}} \simeq \mathbf{C}_{Z-\hat{Z}}$, and (ii) is uncorrelated with the kriging component $\hat{\mathbf{z}}$, i.e.: $\mathbf{C}_{\hat{Z},Z^{(s)}-\hat{Z}^{(s)}} \simeq \mathbf{C}_{\hat{Z},Z-\hat{Z}}$, see Equation (26).

It therefore follows that the covariance matrix $\mathbf{C}_{Z^{(s)}}$ of the simulated vector $\mathbf{z}^{(s)}$ can be written as:

$$\mathbf{C}_{Z^{(s)}} = \mathbf{C}_{\hat{Z}} + \mathbf{C}_{Z^{(s)}-\hat{Z}^{(s)}} \simeq \mathbf{C}_{\hat{Z}} + \mathbf{C}_{Z-\hat{Z}} = \mathbf{C}_{Z} \tag{32}$$

which entails that the conditionally simulated point-support values reproduce approximately (within ergodic fluctuations) the point-support covariance model $C_Z(\mathbf{h})$. Note also that in the multivariate Gaussian case, whereby $\mathbf{d} \sim N(\mathbf{m}_D, \mathbf{C}_D)$, the point-support realizations are also multivariate Gaussian, i.e., $\mathbf{z}^{(s)} \sim N(\mathbf{m}_Z, \mathbf{C}_Z)$.

It is also straightforward to prove that the resulting area-to-point simulated realization $\mathbf{z}^{(s)}$ reproduces the areal-support data, i.e., to prove that $\mathbf{G}\mathbf{z}^{(s)} = \mathbf{d}$. Indeed, from Equation (30):

$$\mathbf{G}\mathbf{z}^{(s)} = \mathbf{G}\hat{\mathbf{z}} + \mathbf{G}\mathbf{z}^{(s)} - \mathbf{G}\mathbf{C}_{ZD}\mathbf{C}_D^{-1}\mathbf{d}^{(s)} \tag{33}$$

Using Equations (20) and (29), as well as the last relation of Equation (12), the above expression becomes:

$$\mathbf{G}\mathbf{z}^{(s)} = \mathbf{d} + \mathbf{d}^{(s)} - \mathbf{C}_D\mathbf{C}_D^{-1}\mathbf{d}^{(s)} = \mathbf{d} \tag{34}$$

which entails that any areal-support datum $d(v_k)$, defined as the convolution of point values within v_k with the sampling function g_k, can be reproduced exactly by the resulting area-to-point simulated realizations, as long as the latter are generated via Equation (30).

It should be stressed here that the coherence characteristic of the simulated point realizations does not require that the vector $\mathbf{d}$ of areal data be Gaussian distributed. The ensemble mean and variance over a large number S of simulated realizations $\{\mathbf{z}_c^{(s)}, s = 1, \ldots, S\}$, however, will vary depending on the multivariate properties of the point-support RF $\{Z(\mathbf{u}_p), p = 1, \ldots, P\}$.

In what follows, we demonstrate the application of area-to-point prediction and stochastic simulation via a case study. Without loss of generality, we use a simulated set of point-support values for reference in order to to evaluate the reproduction of their spatial characteristics from the area-to-point predictions/simulations. All computations were performed in MATLAB using code that was written by the authors.

3. Case Study

The setting of this case study is that of areal data available on a fixed rectangular support, similar to the pixels of remotely sensed imagery. The objective is that of prediction/simulation of point-support values from pixel-support data. In what follows, we use subscripts p and q to denote the row and column indices of the pq-th prediction/simulation location $\mathbf{u}_{pq}$ of a regular 2D grid. The corresponding point-support value is denoted as $z(\mathbf{u}_{pq})$. Similarly, we use subscripts k and l to denote the row and column indices of the kl-th pixel v_{kl} of a regular pixel-support 2D grid. The corresponding pixel-support value is denoted as $d(v_{kl})$.

3.1 Reference point values and sample areal data

The reference point-support values, which are subsequently used to obtain the pixel-support data, are generated via moving average simulation in the frequency domain, see Section 2.3. In this case study, the simulation parameters include a mean $m_Z = 50$ (data units), and an isotropic point semivariogram model:

$$\gamma(\mathbf{h}) = \gamma(|\mathbf{h}|) = C_Z(\mathbf{0})\left[1 - \exp\left(-\frac{3|\mathbf{h}|}{r}\right)\right] = 10\left[1 - \exp\left(-\frac{3|\mathbf{h}|}{100}\right)\right] \tag{35}$$

with variance $C_Z(\mathbf{0}) = 10$ (data units)^2, and practical range (distance at which 95% of the model sill is reached) $r = 100$ (distance units).

The simulated reference point-support values $\{z(\mathbf{u}_{pq}), p, q = 1, \ldots, 594\}$ on a 594×594 regular grid (of unit cell size) are shown in Figure 1A. These reference point values are subsequently averaged using non-overlapping pixels of size 11×11, thus resulting to a 54×54 gridded set of pixel-support data $\{d(v_{kl}), k, l = 1, \ldots, 54\}$ shown in Figure 1B. In other words, the sampling function in this case is a simple box-car filter (Collins and Woodcock, 1999). The histograms of the reference point values and the resulting pixel-support data are shown in Figures 1C-D. Note that the mean of the areal data is the same as that of the point values (49.7), and that the variance of the latter (3.21^2) is larger than that of the former (2.95^2). The semivariogram of the reference point values of Figure 1A, along with the reference model of Equation (35) used for their generation, is shown in Figure 1E. The semivariogram of the areal data of Figure 1B, along with the regularized semivariogram $\overline{\gamma}_Z(\mathbf{h})$, is shown in Figure 1F. The regularized semivariogram is obtained in a similar way as Equation (6), by using a box-car filter as a sampling function (Collins and Woodcock, 1999). Note the close agreement between the semivariogram of the areal values and the regularized model.

In order to speed-up computations and avoid the inversion of a (potentially) large covariance matrix $\mathbf{C}_D$, area-to-point prediction and simulation at any location $\mathbf{u}_{pq}$ within a pixel v_{kl} is performed using $n = 25$ areal data from a 5×5 pixel template centered at v_{kl}. In this case, the vector of 25 neighboring areal data used for prediction/simulation at all 121 points within v_{kl} is denoted as: $\mathbf{d}_{kl} = [d(v_{k+j,l+j}), j = -2, -1, 0, 1, 2]$. In order to avoid edge effects, prediction/simulation is performed at the 550×550 grid nodes of the subregion shown with a dashed line in Figure 1A. Thus, the matrix of grid nodes considered for prediction/simulation is: $[\mathbf{u}_{pq}, p, q = 23, \ldots, 573]$; the first and last 2×11 rows, as well as the first and last 2×11 columns, of the 2D grid of Figure 1A are discarded. All subsequent images of predicted/simulated values pertain to this interior grid. Consequently, the reproduction of the pixel-support data from the area-averaged point predictions/simulations is hereafter investigated only for the subregion of 50×50 pixels shown with a dashed line in Figure 1B. Note, however, that all 54×54 areal data of Figure 1B are used for prediction/simulation.

3.2 Area-to-point kriging

A regular 2D prediction/simulation grid, coupled with gridded areal data considered on a fixed template, leads to great computational savings because the data configuration (relative position of pixels with respect to each other within the template) does not change from the prediction locations within one pixel to another. This entails that the kriging weights and the kriging variance, which do not depend on the data values but only on the data configuration, need only be calculated once (only for a single template).

Area-to-point kriging (see Section 2.2) is performed at all 550×550 grid nodes of the interior region of Figure 1A, using the 54×54 pixel-support data of Figure 1B and the reference point

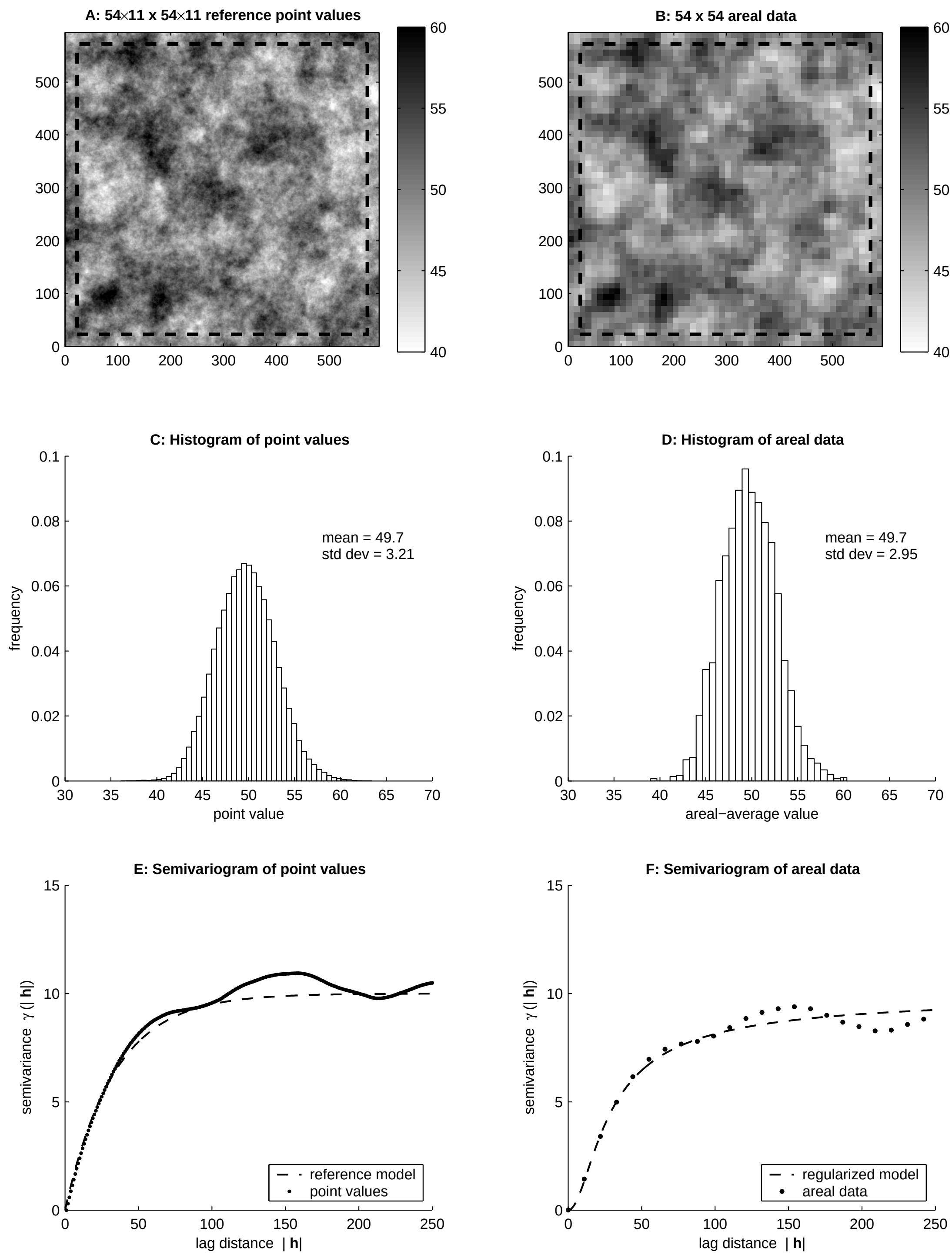

Figure 1: **A:** Reference point values on a 594×594 regular grid of unit cell size generated via Gaussian moving average simulation (see text for details); the dashed-line box indicates the 550×550 subregion where prediction/simulation is performed. **B:** Area-averaged data obtained from (**A**) by averaging over 54×54 non-overlapping pixels (each containing 11×11 point values); the dashed-line box encloses the 50×50 pixel-support data whose reproduction is subsequently evaluated. **C-D:** Histograms of point values and areal data. **E-F:** Semivariograms of point values and areal data, along with the reference point model used for the simulation and its regularized counterpart.

semivariogram model of Equation (35); the resulting point predictions are shown in Figure 2A. Note the similarity of these predictions with the reference point values of the interior region of Figure 1A: the correlation coefficient between these two maps is 0.95. The map of associated standard errors of prediction for the same subregion is shown in Figure 2B. The resulting pattern of standard errors has a mesh-type appearance due precisely to the fact that the kriging variance depends only on the data configuration (and the semivariogram model). In other words, the map of kriging standard errors is comprised of replicates of small matrices of such errors (each of size 11×11). Standard errors are minimum at the pixel centers, and increase radially (due to the point semivariogram model $\gamma_Z(\mathbf{h})$ being isotropic) towards the pixel boundaries. Note, however, that the standard error does not attain the value of zero at the pixel center; this would be the case only if the areal data were assumed of point support.

Figure 2C gives the scatterplot between the area-averaged point predictions and the corresponding pixel-support data in the subregion of Figure 1B, which indicates a perfect reproduction; this is expected, since such a reproduction of pixel-support data is guaranteed by construction, per Equation (20). The quantile-quantile plot between the distributions of point predictions and the reference values is shown in Figure 2D. Note the smoothing effect of interpolation, which leads to overestimation of the proportion of low reference values (lower end of q-q plot) and underestimation of the proportion of high reference values (upper end of q-q plot). The semivariogram of point predictions, along with the reference point semivariogram model of Equation (35) is shown in Figure 2E. Note again the smoothing effect of interpolation, which results into a semivariogram with parabolic behavior near the origin, and lower sill (less variance). The above smoothing effects of interpolation are corrected via stochastic simulation at the expense of reduced local accuracy (lower correlation between point predictions and reference values), see for example Chilès and Delfiner (1999).

We further investigate the effect of the point semivariogram model on the resulting area-to-point predictions and their standard errors. More precisely, we investigate the use of two alternative semivariogram models:

$$\begin{aligned}
\gamma(|\mathbf{h}|) &= 5\left[1-\delta_{|\mathbf{h}|}\right] + 5\left[1 - \exp\left(-\frac{3|\mathbf{h}|}{100}\right)\right] && (36)\\
\gamma(|\mathbf{h}|) &= 10\left[1-\delta_{|\mathbf{h}|}\right] && (37)
\end{aligned}$$

where $\delta_{|\mathbf{h}|}$ denotes the Kroneker delta function, defined as $\delta_{|\mathbf{h}|} = 1$ if $|\mathbf{h}| = 0$ and $\delta_{|\mathbf{h}|} = 0$ if $|\mathbf{h}| \neq 0$. The semivariogram model of Equation (36) is similar to the reference model of Equation (35), but with a 50% relative nugget. Equation (37) is a pure nugget effect model (complete absence of spatial correlation).

Figures 3A-B show the area-to-point predictions using the above two semivariogram models. The predictions of Figure 3A are very similar to (but somewhat less smooth than) those of Figure 2A, while the predictions of Figure 3B are very similar to the pixel-support data of the interior region of Figure 1B. The correlation coefficients between these two sets of point predictions and the reference point values of the interior region of Figure 1A are 0.94 and 0.92, respectively. In both cases, the area-averaged predictions reproduce exactly the pixel-support data of the interior region of Figure 1B. Figures 3C-D show the standard errors associated with the predictions of Figures 3A-B. Note the similar pattern (but higher magnitude due to the higher nugget effect) of the standard errors of Figure 3C with those of Figure 2B. The standard error for the semivariogram model

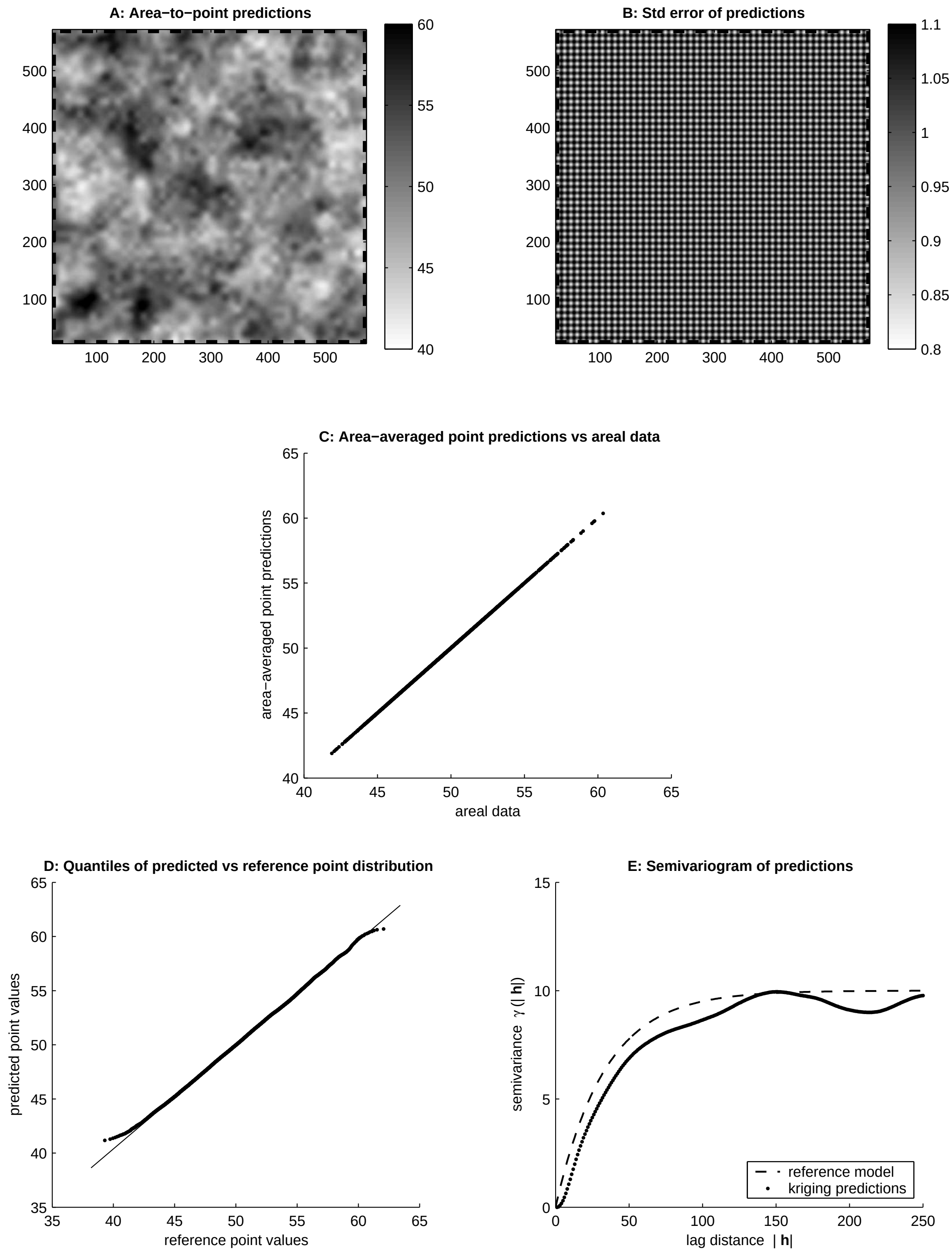

Figure 2: **A:** Simple area-to-point kriging predictions on the 550×550 grid shown in Figure 1A, using the 54×54 pixel-support data of Figure 1B and the reference point semivariogram model (see text for details). **B:** Standard error of area-to-point predictions. **C:** Scatterplot of area-averaged point predictions versus the corresponding 50×50 areal data of Figure1B. **D:** Quantile-quantile plot between the distributions of predictions and reference point values. **E:** Semivariogram of point predictions, along with the reference point model.

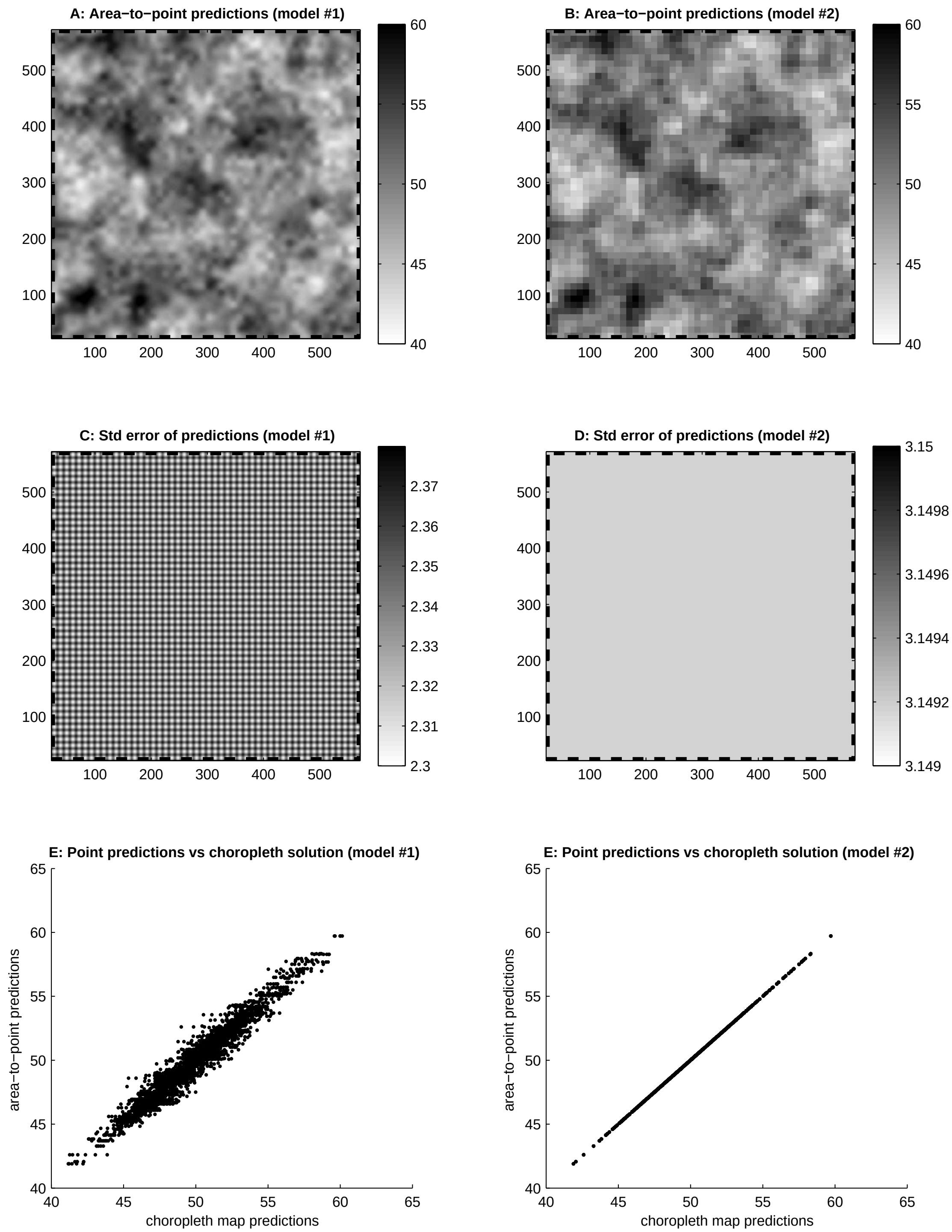

Figure 3: **A:** Simple area-to-point kriging predictions on the 550×550 grid shown in Figure 1A, using the 54×54 pixel-support data of Figure 1B and the reference point semivariogram model with 50% relative nugget (see text for details). **B:** Same as (**A**), but using a pure nugget effect point semivariogram model. **C-D:** Standard error of area-to-point predictions for cases (**A**) and (**B**). **E-F:** Scatterplots of area-to-point predictions for cases (**A**) and (**B**) versus those corresponding to the choropleth map solution (see text for details).

of Equation (37) is a constant, see Figure 3D, and approximately equal to the sill $\overline{\gamma}_Z(\infty)$ of the regularized semivariogram (Journel and Huijbregts, 1978); this is expected, since the information content of the block data in this case is nill.

Figures 3E-F give the scatterplots of the area-averaged point predictions under these two semivariogram models versus the predictions obtained under the choropleth map solution of Equation (21), whereby the point values within each pixel v_{kl} are all assumed the same and equal to the corresponding areal datum $d(v_{kl})$. As the nugget effect increases the area-to-point predictions become less smooth and more similar to the choropleth map solution. At the limiting case of a pure nugget effect point semivariogram model, the area-to-point predictions identify those obtained via the choropleth map solution. The implications of this result are addressed in the discussion section.

3.3 Area-to-point stochastic simulation

Area-to-point stochastic simulation (see Section 2.3) is performed at all 550×550 grid nodes of the interior region of Figure 1A, using the 54×54 areal data of Figure 1B and the reference point semivariogram model of Equation (35). Figures 4A-B show two such area-to-point Gaussian realizations. The two realizations appear very similar, due to the influence of the areal data, but they are different; their difference is shown in Figure 4F. The main spatial features of the reference point values of Figure 1A are well captured: the correlation coefficient between the simulated and reference values is 0.90 in both cases. This correlation is lower than the corresponding one derived from the kriging predictions, due to the reduced local accuracy of stochastic simulation when compared to kriging. Note, however, that such a reduced local accuracy is compensated by enhanced distributional and structural accuracy, i.e., closer reproduction of the histogram and semivariogram model inferred from the available data. Figure 4C gives the quantile-quantile plots between the distributions of simulated and reference point values; the close reproduction of the reference distribution is evident. Figure 4D gives the semivariograms of the two simulated realizations, along with the reference point model of Equation (35); the close semivariogram reproduction is again evident. Last, Figure 4E shows the exact reproduction of the 50×50 pixel-support data of Figure 1B from the two simulated realizations, a characteristic that is ensured per the use of kriging in the simulation procedure, see Equation (34).

We conclude by investigating the use of semivariogram model of Equation (36) in a simulation setting. This semivariogram model has a 50% relative nugget, and is thus inconsistent with the reference point model of Equation (35) that was used to generate the reference point values and consequently the pixel-support data of Figure 1B.

Figures 5A-B show two area-to-point Gaussian realizations for the interior region of Figure 1A, generated using the 54×54 areal data of Figure 1B and the point semivariogram model of Equation (36). The two realizations appear more noisy than the two realizations of Figures 4A-B due to the high relative nugget effect. The difference between these two realizations is shown in Figure 5F. Note that the spatial features of the reference point values of Figure 1A are not captured as well (the correlation coefficient between simulated and reference values is 0.75 in both cases). Figure 5C gives the quantile-quantile plots between the distributions of simulated and reference point values; the reproduction of the reference distribution is less perfect than that of Figure 4C.

Figure 5D gives the semivariograms of the two simulated realizations, along with the reference point model of Equation (35) and the model of Equation (36) that was used for the simulation.

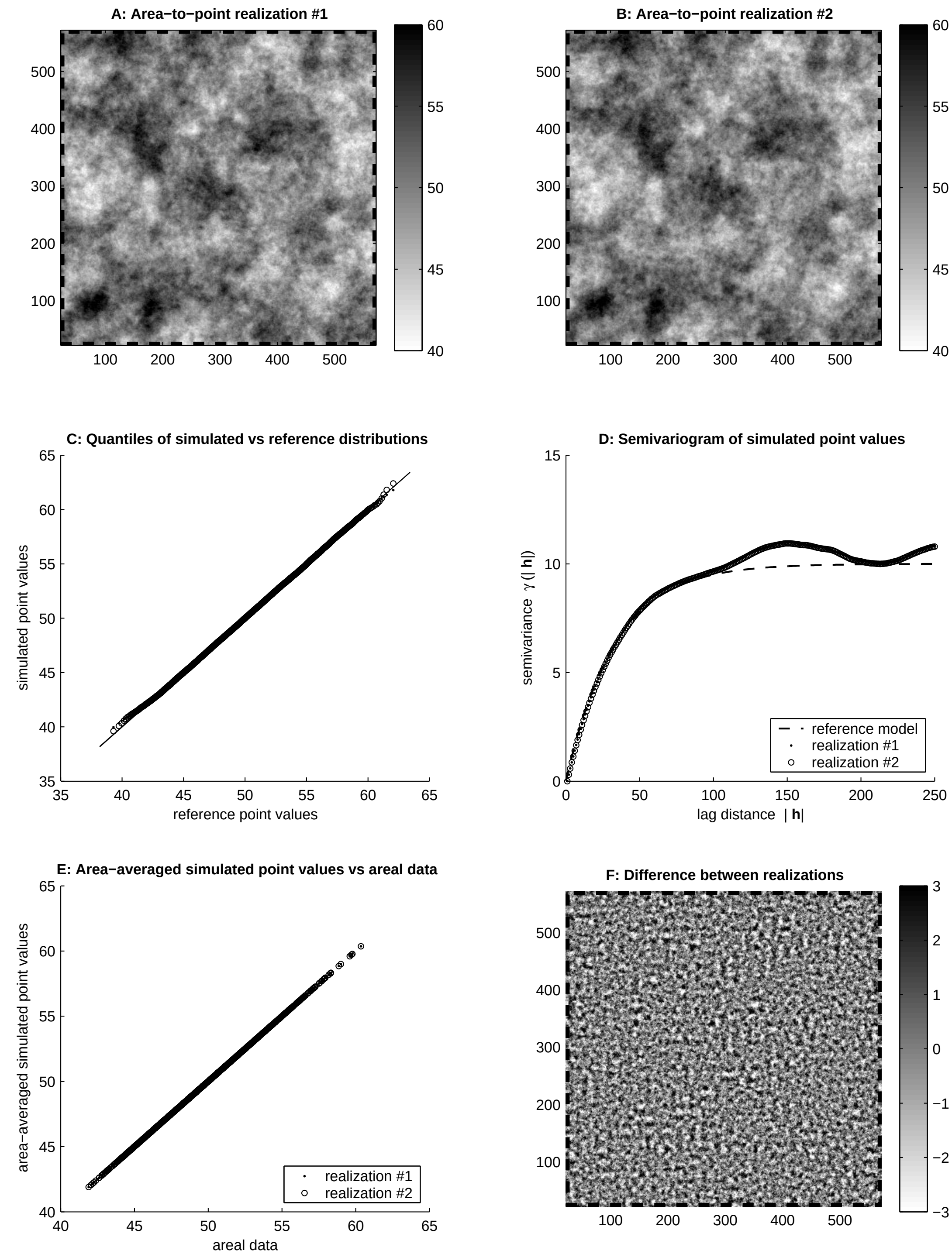

Figure 4: **A-B:** Two area-to-point Gaussian realizations on the 550×550 grid shown in Figure 1A, generated using the 54×54 pixel-support data of Figure 1B and the reference point semivariogram model (see text for details). **C:** Quantile-quantile plot between the distributions of simulated and reference point values. **D:** Semivariograms of simulated realizations, along with the reference point model. **E:** Scatterplot of area-averaged simulated point values versus the corresponding 50×50 pixel-support data of Figure1B. **F:** Difference between realizations (**A**) and (**B**).

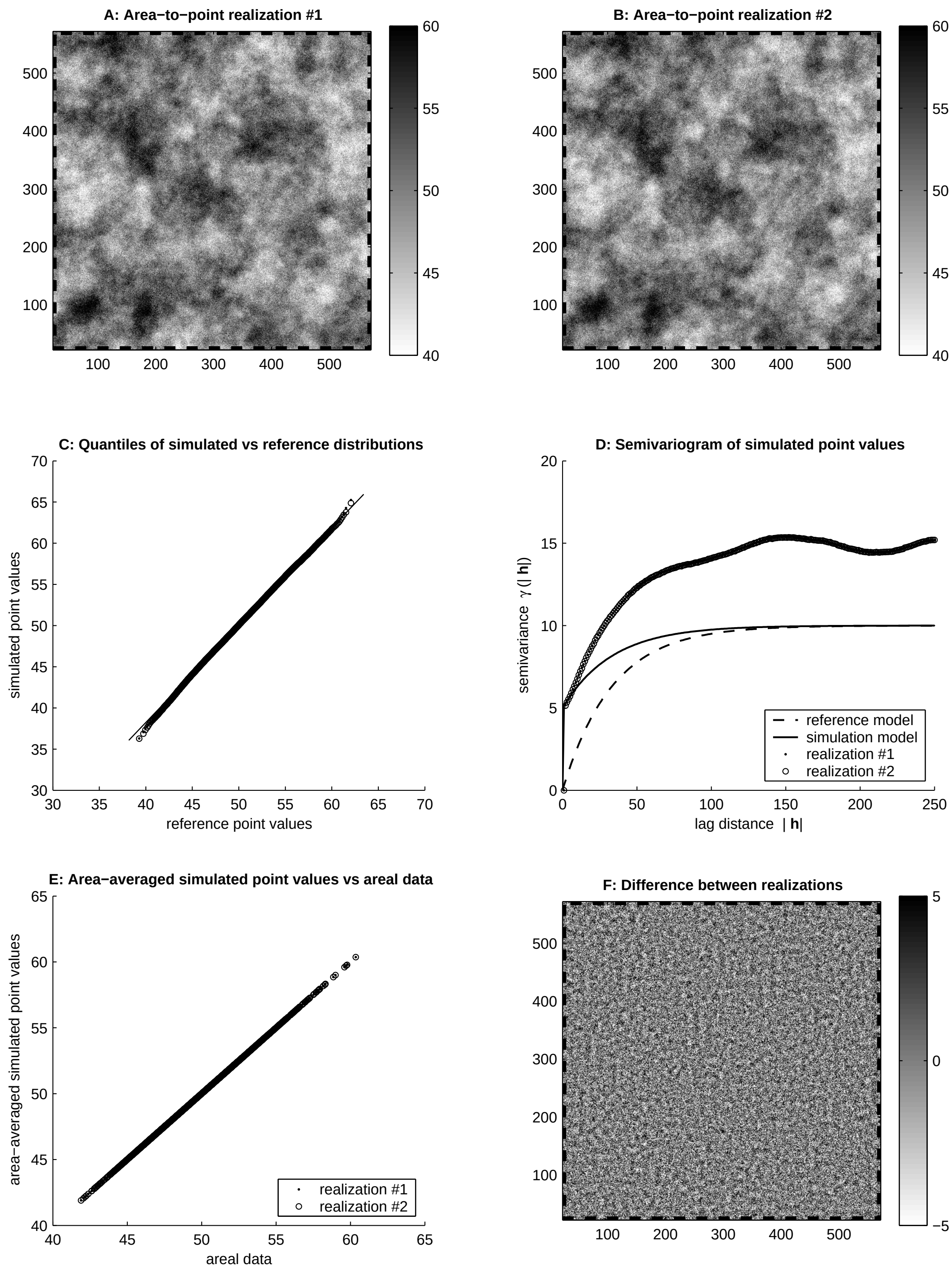

Figure 5: **A-B:** Two area-to-point Gaussian realizations on the 550×550 grid shown in Figure 1A, generated using the 54×54 pixel-support data of Figure 1B and the reference point semivariogram model with 50% relative nugget (see text for details). **C:** Quantile-quantile plot between the distributions of simulated and reference point values. **D:** Semivariograms of simulated realizations, along with the reference point model and that used for simulation. **E:** Scatterplot of area-averaged simulated point values versus the corresponding 50×50 pixel-support data of Figure1B. **F:** Difference between realizations (**A**) and (**B**).

Interestingly enough, the semivariograms of the simulated realizations appear to be shifted versions of the reference model by an amount approximately equal to the relative nugget of the model used for the simulation. Any inconsistency between the point semivariogram that is derived from the deconvolution of the semivariogram of the areal data (the reference point model in this case) and the model used for simulation, leads to erroneous semivariograms for the simulated realizations. Last, Figure 5E shows the exact reproduction of the 50×50 pixel-support data of Figure 1B from the two simulated realizations, a characteristic that is again ensured per the use of kriging in the simulation procedure.

4. Discussion

In this paper, a geostatistical approach has been developed for obtaining point-support predictions and simulated realizations from areal data, the latter defined as convolutions of point-support values within the support of each areal datum with a sampling kernel. Such a general definition accounts for areal data obtained as weighted linear combinations of point values within an areal support, and includes summation, averaging, and differentiation as particular cases.

It has been proved that, given a point-support covariance model, consistent computation of all auto- and cross-covariances between areal data and point values leads to coherent predictions/simulations. In other words, the areal data are reproduced by the resulting point-support predictions/simulations when the latter are convolved with the sampling kernel used to define the areal data. The simulated point-support realizations also reproduce the (Gaussian in this work) histogram and the semivariogram of the point-support values. A case study using areal data defined as pixel-support averages of (simulated) point values is also presented, in order to illustrate the application of the proposed methodology in a realistic setting.

It has also been demonstrated that the choropleth map solution corresponds to the extremely particular, and in most cases unrealistic, assumption of spatially uncorrelated point-support values. This result bears important implications for the current practice in geography, whereby the use of the choropleth map is justified by invoking the assumption of constant point values within any areal support. This assumption is often easier to defend than the assumption of uncorrelated point-support values, but it is this latter assumption that is a prerequisite for the former.

On-going research is focused on: (i) extending the proposed framework to accommodate non-Gaussian data, (ii) accounting for areal data that are not error-free and consequently should not be reproduced exactly, (iii) inferring the point-support covariance/semivariogram model from the available areal data via deconvolution in the frequency domain, and (iv) integrating point-support and areal data using cokriging. Further case studies using (real) data defined as other types of convolutions, e.g., data defined as derivatives of point values, should be conducted to investigate the robustness of the proposed methodology with respect to areal data type and complexity of spatial characteristics (e.g., nested and anisotropic covariance structures).

5. References

Anderson, T., 1958. *An Introduction to Multivariate Statistical Analysis* (New York: John Wiley & Sons).

Atkinson, P., and Tate, N., 2000, Spatial scale problems and geostatistical solutions: A review. *Professional Geographer*, **52**, 607–623.

Chilès, J.-P., and Delfiner, P., 1999. *Geostatistics: Modeling Spatial Uncertainty* (New York: John Wiley & Sons).

Collins, J., and Woodcock, C., 1999, Geostatistical estimation of resolution-dependent variance in remotely sensed images. *Photogrammetric Engineering & Remote Sensing*, **65**, 41–50.

Deutsch, C., and Journel, A., 1998. *GSLIB: Geostatistical Software Library and User's Guide*, Second Edition (New York: Oxford University Press).

Gotway, C., and Young, L., 2002, Combining incompatible spatial data. *Journal of the American Statistical Association*, **97**, 632–648.

Journel, A., and Huijbregts, C., 1978. *Mining Geostatistics* (New York: Academic Press).

Kyriakidis, P., 2003, The geostatistical solution of the area-to-point spatial interpolation problem. *Geographical Analysis*, In press.

Lam, N., 1983, Spatial interpolation methods: A review. *The American Cartographer*, **10**, 129–149.

Le Ravalec, M., Noetinger, B., and Hu, L., 2000, The FFT moving average (FFT-MA) generator: An efficient numerical method for generating and conditioning Gaussian simulations. *Mathematical Geology*, **32**, 701–723.

Matheron, G., 1971. *The Theory of Regionalized Variables and its Applications* (Paris: Fasc. 5, Ecole Nat. Sup. des Mines).

Menke, W., 1989. *Geophysical Data Analysis: Discrete Inverse Theory*, Revised Edition (San Diego: Academic Press).

Oliver, D., 1995, Moving averages for Gaussian simulation in two and three dimensions. *Mathematical Geology*, **27**, 939–960.

Tobler, W., 1979, Smooth pycnophylactic interpolation for geographical regions. *Journal of the American Statistical Association*, **74**, 519–536.

Vanmarcke, E., 1983. *Random Fields: Analysis and Synthesis* (Cambridge, MA: The M.I.T. Press).